

Java Fundamentals & Control Flow - Quick Revision Cheat Sheet

QUICK REVISION & EXAM CHEAT SHEET

Subject: Java Programming | Chapter: Chapter 1: Java Fundamentals & Control Structures | Academic Year: 2026-2027

Core Architecture Summary

- * JDK = JRE + Development Tools (compiler, debugger). JRE = JVM + Standard Core Libraries.
- * Bytecode (.class) is executed by the JVM interpreter and compiled to native machine code via JIT.
- * Java is strictly Pass-by-Value for both primitive types and object references.

Data Types & Memory Sizes

- * byte: 1 byte (-128 to 127) | short: 2 bytes | int: 4 bytes | long: 8 bytes (suffix L)
- * float: 4 bytes (suffix f) | double: 8 bytes (default decimal) | char: 2 bytes (Unicode) | boolean: 1 bit logical
- * Widening conversion is automatic; Narrowing requires explicit cast `(targetType) value`.

Control Flow Key Rules

- * switch works with byte, short, char, int, String, and Enums. Floating point types are NOT supported.
- * do-while executes the body at least once regardless of condition validity.
- * Enhanced for loop syntax: `for (ElementType item : collection)`.

Important Exam Traps & Best Practices

- * Variables declared inside a method (local variables) must be initialized before use; they receive no default values.
- * Instance variables and static fields are automatically initialized to default values (0, null, false).
- * String objects in Java are immutable; string concatenation inside intensive loops should use StringBuilder.